Research Statement
CIFRE PhD — CREST - IMT, Generali France

Regime-Conditional Factor Dynamics, Functional Capital Proxies and Optimal Stochastic Control for Insurance Portfolio Management

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Thesis in one sentence.

> *This thesis introduces the first* regime-conditional generative framework with certified W_2 guarantees under online regime inference*, coupled to a* differentiable functional capital proxy*, to enable a fully dynamic, capital-aware Total Portfolio Approach for a life insurer under Solvency II. The three axes form a single closed control loop : a certified scenario generator (Axis 1) feeds a fast differentiable SCR proxy (Axis 2), which becomes the capital constraint of a joint allocation-and-hedging policy (Axis 3).*

1. INDUSTRIAL CONTEXT AND MOTIVATION

1.1. The 2022 dependence crisis and the limits of static allocation

The year 2022 was a structural stress test for insurance asset managers : unlike earlier single-asset shocks, equities and bonds fell simultaneously and diversification broke down. For a life insurer such as Generali France, this exposed a structural flaw in Strategic Asset Allocation (SAA), which assumes stable, linear dependence between economic factors. The equity/bond correlation switched from -0.3 (diversifying) to $+0.6$ (concentration risk) within weeks — a regime reversal that fixed-correlation models cannot anticipate.

This motivates the central problem of this thesis : *how to characterise and optimally control the solvency position of an insurer when the joint dynamics of economic factors are governed by a latent Markov chain inducing non-stationary, non-linear dependence structures across regimes ?*

1.2. Three compounded bottlenecks in the current SAA process

The SAA process at Generali France runs in three sequential steps, each with structural limitations. **(1) Economic Scenario Generator (ESG)** — a proprietary ESG produces thousands of trajectories (rates, spreads, equity, real estate, inflation) calibrated with **fixed correlations averaged across regimes**, feeding all downstream actuarial calculations. **(2) SCR via nested LSMC** — the Solvency Capital Requirement, the 99.5% one-year VaR of net asset value change, requires the Best Estimate of Liabilities (BEL) under 10,000+ scenarios ; Least-Squares Monte Carlo (LSMC, Krah et al., 2018) replaces the inner simulation by an offline polynomial proxy. **(3) Annual Markowitz optimisation** — the SCR is a fixed constraint in a mean-variance optimisation revised **once per year**, with no automatic identification of the dominant portfolio objective (duration, inflation, carry, capital efficiency), even though the relevant factor varies across portfolios and regimes.

Four structural limitations.

1. **Regime-blind dependence.** Fixed correlations cannot anticipate regime reversals : a Markowitz optimum for a low-inflation regime becomes high-risk under stagflation.
2. **LSMC proxy limitations.** The polynomial proxy is accurate only within its regime-agnostic calibration region, degrading precisely in the stress zones most relevant for capital decisions, and is not expressed in common economic factors.
3. **Static, siloed decision-making.** Allocation and hedging are decided separately — suboptimal for an insurer with convex critical losses — and the objective is imposed uniformly rather than inferred from each portfolio's dominant risk factor.
4. **Private asset valuation bias.** Private equity, private debt and real estate NAVs are smoothed quarterly : in 2008 private equity NAVs stayed stable for 6–12 months before correcting sharply, understating stress-regime dependence and introducing a lag bias in the SCR.

1.3. The Total Portfolio Approach : target framework

The **Total Portfolio Approach (TPA)** evaluates each decision through its marginal contribution to total balance-sheet risk via common factors (Duration, Inflation, Growth, Liquidity, Carry). Implementations at sovereign wealth funds (Future Fund, CPP Investments) report about +1% excess return per year over classical SAA (Thinking Ahead Institute, 2025 ; CAIA, 2024). For an insurer under Solvency II, the factor space must also include liability-side factors (surrender, longevity, yield-curve/BEL interactions), and every decision must internalise a capital cost. Existing actuarial tools address none of this jointly ; this thesis fills the gap along three coupled axes : regime-conditional factor simulation (Axis 1), fast capital proxy (Axis 2), and joint allocation-hedging policy (Axis 3).

2. RESEARCH OVERVIEW

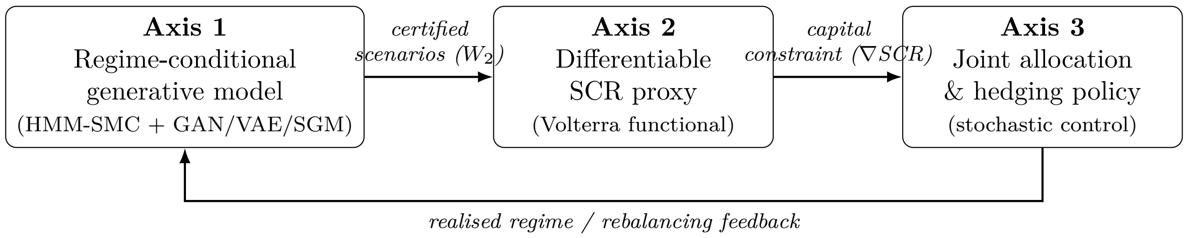

FIGURE 1 – The three axes form a single closed capital-aware control loop : certified regime-conditional scenarios feed a fast differentiable capital proxy, which becomes the explicit constraint of the joint allocation and hedging policy ; realised regimes feed back into the generator.

Approach	Regime-aware	Differentiable	W_2/L^2 certified	Insurance-calibrated
Static SAA / Markowitz	✗	✗	✗	✓
Copulas (Sklar, Fermanian)	✗	✗	✓	✗
LSMC proxy (Krah et al.)	✗	✗	✓	✓
GAN/VAE time-series (Wiese, Yoon)	✗	✓	✗	✗
Score-based models (Ocello et al.)	✗	✓	✓	✗
Deep hedging (Buehler et al.)	✗	✓	✗	✗
This thesis (Axes 1–3)	✓	✓	✓	✓

TABLE 1 – Positioning of the research programme against the state of the art.
✓ = requirement met, ✗ = not addressed.

3. RESEARCH PROGRAMME

3.1. Axis 1 — Regime-Conditional Factor Dependence

Research question.

Research Question 1. *How to generate the joint distribution of TPA factors conditionally on a sequentially estimated macroeconomic regime, while providing : (i) formal goodness-of-fit guarantees on the learned dependence structure ; and (ii) non-asymptotic W_2-convergence bounds under particle-filter inference error on the regime state ?*

State of the art.

Regime identification. Hidden Markov Models (Hamilton, 1989) and Sequential Monte Carlo (SMC) particle filters (Chopin & Papaspiliopoulos, 2020) provide tractable real-time estimation of latent macroeconomic regimes $z_t \in \{1, \ldots, K\}$. Their combination — an online HMM-SMC filter — yields a posterior $p(z_t|\mathcal{F}_t)$ updated at each observation.

Generative Adversarial Networks. Applied to financial series by Wiese et al. (2020, Quant-GAN) and Yoon et al. (2019, TimeGAN), conditional GANs generate $\mathbf{F}_{t+1}|z_t = k \sim G_\theta(\varepsilon, k)$ with full flexibility (no parametric dependence assumption), but suffer from *mode collapse* and the absence of distributional guarantees, limiting regulatory auditability.

Variational Autoencoders. CVAEs encode each trajectory into a regime-conditional posterior and maximise the ELBO ; they offer stable training and an auditable latent structure, but produce overly smooth samples due to the Gaussian posterior approximation.

Score-Based Generative Models (SGMs). Strasman, Ocello et al. (TMLR, 2024) establish W_2 convergence bounds :

$$\mathrm{KL}(\hat{p} \,\|\, p^*) \;\leq\; \int_0^T g(t)^2 \,\varepsilon_{\mathrm{score}}(t)\, dt + \varepsilon_{\mathrm{init}}. \tag{1}$$

Gentiloni-Silveri & Ocello (ICML, 2025) extend these guarantees to non-log-concave distributions (Gaussian mixtures), directly applicable to multi-regime market models. De Marco, Pham & Zanni (2025) further introduce jump-diffusion Schrödinger bridges to capture discontinuous regime transitions.

Copulas : a validation framework, not a generative model. Sklar's theorem separates margins from dependence, and Fermanian (2013, 2022) provides goodness-of-fit tests for conditional copulas. Classical parametric families (Gaussian, Student-t, Archimedean), however, impose rigid tails, scale poorly in high dimension and cannot capture cross-regime dynamics. Copulas therefore serve here exclusively as a *validation tool* : the conditional tail-dependence coefficient λ_k is estimated and tested via Fermanian's specification tests to validate generative outputs, not to generate them.

Financial networks and correlation structure. Bardoscia et al. (2021) show that asset correlation matrices exhibit community structure and spectral properties amenable to graph-theoretic analysis, motivating an optional graph-neural-network (GNN) encoder in the Axis 1 generator to capture non-linear pairwise interactions beyond the covariance matrix.

Identified gaps. Neither conditional GANs, VAEs nor SGMs have been applied to the joint distribution of TPA factors with real-time regime identification. For the GAN/VAE : the absence of formal regime-conditional tail-dependence validation. For the SGM : the absence of W_2 bounds for the conditional case with sequentially estimated regime states. These constitute the theoretical contribution of Axis 1.

Proposed approach.

Stage 1 (Year 1) — cGAN and cVAE. Let z_t be identified by an online HMM-SMC filter. A conditional GAN $G_\theta(\varepsilon, k)$ and a CVAE are trained in parallel, both incorporating the private asset vintage lag factor $\tilde{F}_t^{\mathrm{PA}} = \mathrm{NAV}_{t-1}^{\mathrm{PA}}$ to capture the quarterly smoothing bias. Model comparison is conducted on : Wasserstein-2 distance W_2, Maximum Mean Discrepancy (MMD), per-regime tail calibration via Fermanian's λ_k tests, autocorrelation structure, and regulatory auditability.

Stage 2 (Year 2) — Conditional SGM with W_2 guarantees. A conditional OU forward process for regime k :

$$d\mathbf{x}^\tau = -\mathbf{x}^\tau \, d\tau + \sqrt{2}\, g_k(\tau)\, dW_\tau, \quad \tau \in [0, T], \tag{2}$$

with regime-specific noise schedule $g_k(\tau)$. The theoretical contribution is the extension of the W_2 bounds of Gentiloni-Silveri & Ocello (ICML 2025) to the conditional case with online SMC regime estimation — quantifying how inference error in $p(z_t|\mathcal{F}_t)$ propagates into the generative bound. A scenario selection layer then extracts a central regime-conditional scenario for downstream reporting and optimisation diagnostics.

3.2. Axis 2 — Supervised Capital Proxy

Research question.

Research Question 2. *How to construct a proxy for the insurance SCR — expressed as a function of TPA factors, the current regime, and liability states including private asset NAV lags formalised as a Volterra functional — that is accurate in decision-relevant zones with controlled L^2 error bounds per regime and sub-10ms inference time, and exposes a differentiable interface for its direct embedding as an explicit capital constraint in a downstream stochastic optimisation programme ?*

State of the art.

Classical proxy models. The LSMC framework (Krah et al., 2018) approximates the BEL by polynomial regression on a design-of-experiment. Its two critical limitations are regime-blindness and degraded accuracy in stress zones. Neural network proxies (Huber & Schmocker, 2022) achieve 1–3% accuracy at < 10ms inference. Bourgey (2020) provides theoretical foundations via multi-level Monte Carlo (MLMC) estimators for nested expectations and polynomial chaos expansions with L^2 error estimates, motivating the adaptive-weighting architecture of Axis 2.

Recent generative and flow-based surrogates. Three recent architectures merit systematic comparison for their uncertainty-quantification and stability properties : **Neural ODEs** (Chen et al., 2018), which model the proxy surface as a continuous-depth dynamical system ; **Conditional Flow Matching** (Lipman et al., 2022), which learns straight-line ODE paths offering stable training and a full conditional density $p(\widehat{\mathrm{SCR}}|\mathbf{F}, z, \mathbf{P})$ for uncertainty-aware allocation ; and **score-based diffusion surrogates**. **Tabular diffusion models** (TabDDPM, Kotelnikov et al., 2023) additionally serve as design-of-experiment augmentation for under-represented regimes.

Volterra processes and path-dependent risk measures. The private asset NAV lag is not a Markovian state variable. The observed smoothed value $\widehat{\mathrm{NAV}}_t = \int_0^t K(t,s)\,\mathrm{NAV}^*_s\, ds$ is a Volterra functional of the true (unobservable) NAV process, where $K(t,s)$ is a smoothing kernel. The theory of affine Volterra processes (Abi Jaber, Larsson & Pulido, 2019 ; El Euch & Rosenbaum, 2019) provides a tractable framework for such path-dependent dynamics. This motivates extending the capital risk measure to the functional setting :

$$\rho^V(L_T) = \phi\left(\int_0^T K(T,s)\, \mathrm{CVaR}_\alpha(L_s)\, ds\right), \tag{3}$$

where ϕ is a monetary risk measure. Equation (3) formalises the temporally diffused risk crystallisation of private assets and provides a theoretical grounding for the vintage lag correction embedded in the proxy state vector.

Model comparison and interoperability metrics. Proxy architectures will be evaluated on two sets of criteria. *Comparison metrics* : (i) accuracy — RMSE, relative error at the SCR regulatory threshold, L^∞ norm in stress zones ; (ii) calibration — prediction-interval coverage, reliability diagrams ; (iii) speed — inference latency (target < 5ms) and batch throughput ; (iv) interpretability — SHAP values, integrated gradients, first-order Sobol sensitivity indices $S_i = \mathrm{Var}[\mathbb{E}[\widehat{\mathrm{SCR}}|\mathbf{F}_i]]/\mathrm{Var}[\widehat{\mathrm{SCR}}]$. *Interoperability metrics* : (i) differentiability — $\nabla_{\mathbf{F}}\widehat{\mathrm{SCR}}$ available via automatic differentiation for gradient-based embedding in the Axis 3 optimiser ; (ii) API compatibility with GPS and PyTorch/JAX optimisation frameworks ;

(iii) monotonicity and convexity preservation as required by Solvency II constraints ; (iv) memory footprint for production deployment.

Identified gaps. No existing proxy is jointly conditioned on the macroeconomic regime, TPA factors and private asset NAV lag. None accounts for the path-dependent Volterra structure of NAV smoothing, and none provides the differentiable interface required for its direct embedding as an explicit capital constraint in a dynamic optimisation programme.

Proposed approach. We learn the supervised proxy :

$$\widehat{\mathrm{SCR}}(t) = f_\theta(\mathbf{F}_t,\, z_t,\, \mathbf{P}_t), \tag{4}$$

where $\mathbf{P}_t$ includes effective duration, average guaranteed rate, observed lapse ratio and the vintage lag $\tilde{F}_{t-1}^{\mathrm{PA}}$. The network f_θ is trained with **adaptive sample weighting** concentrating mass near the regulatory threshold and in stress regimes. The generalisation bound :

$$\left|\mathrm{SCR}(t) - f_\theta(\mathbf{F}_t, z_t, \mathbf{P}_t)\right| \;\leq\; \varepsilon_k(n_k, \delta), \tag{5}$$

decreasing with n_k (training points in regime k) and with confidence $1 - \delta$. The four architectures (MLP+attention, Neural ODE, Flow Matching, score-based) are compared on the full metric set above, and the selected proxy is exported as a differentiable module ($\nabla_{\mathbf{F}}\widehat{\mathrm{SCR}}$ available) for direct use as an explicit constraint in the Axis 3 programme.

3.3. Axis 3 — Stochastic Control : Joint Allocation and Deep Hedging

Research question.

Research Question 3. *How to learn a joint stochastic control policy optimising TPA factor exposures and hedging positions under a differentiable SCR capital constraint, with regime-adaptive objectives automatically derived from the dominant risk factor ?*

State of the art.

Deep reinforcement learning for ALM. Wekwete (2023) demonstrated the applicability of Deep RL (PPO, SAC) to insurance ALM in stochastic environments, establishing the feasibility of learning allocation policies in high-dimensional state spaces under regulatory constraints.

Deep hedging. Buehler et al. (2019) introduced deep hedging : learning a hedging strategy via a neural network minimising a convex risk measure without market completeness assumptions. The approach has been extended to extreme risk (Carbonneau, 2021 ; Cao et al., 2023) and multi-asset portfolios under stress regimes, and provides the algorithmic backbone for the joint allocation-hedging objective of Axis 3.

Stochastic control under dynamic risk constraints. Réveillac (2013) studies CRRA utility maximisation subject to dynamic risk constraints on trading strategies, characterised by quadratic BSDEs. Within the class of time-consistent distortion risk measures a three-fund separation result holds — directly informing the stochastic control formulation where the joint policy operates under a dynamic SCR constraint generated by a general class of risk measures.

Shortfall systemic risk measures and capital allocation. Rosazza Gianin et al. (2024) develop a theory of shortfall systemic risk measures with axiomatic capital allocation rules via a generalised collapse-to-the-mean framework. Their results connect to the distortion-based capital constraint of Axis 3 and motivate the use of $\mathrm{CVaR}_{0.99}$ of the capital deficit as the hedging objective.

Identified gaps. No existing framework jointly learns allocation and hedging under : (i) a regime-conditional, non-linear dependence model ; (ii) a fast, differentiable capital constraint ; (iii) private asset engagement restrictions ; and (iv) an automatically identified portfolio-specific objective function. The full coupling of Axes 1 and 2 into a stochastic control loop is the contribution of Axis 3.

Proposed approach. A latent objective map $\psi_t = g_\eta(\mathbf{F}_t, z_t, \mathbf{P}_t, \widehat{\mathrm{SCR}}_t)$ identifies the dominant factor objective per portfolio and regime. The agent state is $s_t = (\mathbf{F}_t, \hat{C}_{z_t}, z_t, \mathbf{P}_t, \widehat{\mathrm{SCR}}_t)$ and the action $(\mathbf{w}_t, \mathbf{h}_t)$

is the joint allocation-hedging pair. The policy π solves

$$\max_{\pi} \mathbb{E}_{\pi}\left[\sum_{t=0}^{T} r_{\psi_t}(\mathbf{w}_t, \mathbf{h}_t, \mathbf{F}_{t+1}) - \mathcal{P}_t\right] \quad \text{s.t.} \quad \widehat{\mathrm{SCR}}_t \leq \bar{C}, \tag{6}$$

where r_{ψ_t} is the regime-and-portfolio-specific objective and $\mathcal{P}_t$ aggregates hedging cost, turnover and new-illiquid-commitment penalties. The hedging component targets $\mathrm{CVaR}_{0.99}$ reduction of the capital deficit, with the regime-conditional factor dependence of Axis 1 embedded directly in the simulation environment.

4. SCIENTIFIC RISKS AND MITIGATION

The programme is ambitious ; its main scientific and operational risks have been anticipated, and each axis is designed with a fallback that preserves the overall contribution.

Risk	**Lik.**	**Mitigation / fallback**
cGAN mode collapse or unstable training (Axis 1)	Med.	The CVAE and conditional SGM are trained in parallel as substitutes ; the SGM enjoys formal W_2 guarantees and is the primary fallback if adversarial training fails.
Conditional W_2 bound not provable in full generality (Axis 1)	Med.	A partial result under piecewise log-concavity (Gaussian-mixture regimes, following Gentiloni-Silveri & Ocello, 2025) already covers the practically relevant multi-regime case.
Limited access to granular BEL / private-asset data	Med.	The actuarial design-of-experiment is augmented synthetically via TabDDPM ; the proxy is validated on historical crises (2008, 2020, 2022).
Proxy accuracy degrades in extreme stress zones (Axis 2)	Low	Adaptive sample weighting concentrates training on decision-relevant tails ; MLMC control variates (Bourgey, 2020) bound the residual L^2 error.
RL policy instability / non-convergence (Axis 3)	Med.	The differentiable proxy enables direct gradient-based control as a fallback to model-free RL ; deep hedging provides a validated warm start.

5. TIMELINE (36 MONTHS)

— **Year 1 — Axis 1.** Bloomberg factor database with NAV lags ; online HMM-SMC calibration ; regime-conditional GAN/CVAE validated on 2008/2020/2022 crises. *Submission : Journal of Multivariate Analysis / Finance & Stochastics.*

— **Year 2 — Axis 2.** Actuarial design-of-experiment (Prophet/MoSes) with TabDDPM augmentation ; SCR proxy architecture comparison and Volterra L^2 bounds ; differentiable proxy embedded as optimisation constraint. *Submission : Insurance : Mathematics and Economics.*

— **Year 3 — Axis 3.** Joint allocation-and-hedging agent with automatic objective identification ; full three-axis coupling ; multi-regime out-of-sample backtesting ; defence. *Submission : Quantitative Finance / JFQA.*

6. CONCLUSION

This thesis delivers a tightly coupled three-module system designed to make the **Total Portfolio Approach operational for a life insurer under Solvency II**, with direct integration into Generali

France's proprietary platform GPS :

— **Module 1 (Axis 1)** replaces static ESG correlations with an auditable regime-conditional generative model, enabling real-time scenario generation that reflects the actual dependence structure of the prevailing macroeconomic regime, including the path-dependent correction dynamics of private assets.

— **Module 2 (Axis 2)** replaces the multi-hour SCR calculation with a < 5ms differentiable proxy grounded in Volterra functional theory, making dynamic portfolio optimisation loops computationally feasible and the capital constraint analytically tractable as an explicit gradient-available term.

— **Module 3 (Axis 3)** replaces the annual, siloed allocation process with a dynamic joint policy that simultaneously optimises TPA factor exposures and hedging positions under an automatically identified portfolio-specific objective and an explicit SCR constraint.

Beyond the industrial deliverables, the thesis makes three mathematically self-contained contributions : the extension of W_2 convergence theory to regime-conditional score-based models under sequential estimation error ; the formalisation of private asset valuation bias through the theory of Volterra processes and path-dependent risk measures ; and the characterisation of optimal stochastic control policies under dynamic, non-Markovian capital constraints derived from a general class of shortfall risk measures. Together, these contributions establish rigorous foundations for a new generation of quantitative tools in insurance balance sheet management — tools that are simultaneously *theoretically grounded*, *computationally efficient*, *interoperable* and *regulatory-compliant*.

Beyond the three years. The certified-generation / differentiable-proxy / stochastic-control triptych is deliberately generic. The same architecture extends naturally to dynamic ORSA projection, climate stress-testing under regime-dependent transition scenarios, and IFRS 17 risk-adjustment computation — opening a longer-term research programme on *certified, capital-aware decision systems* for the insurance balance sheet.

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